

Sharpness of the multi-row containment at $\tau + 1$:

$$B^{(\lambda)} \cap E_{\tau+1}^-|_{V_\lambda} = 0$$

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Abstract

Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$. The Pillar 1 meta-theorem (15 May 2026) proves that the column-1 chain image $B^{(\lambda)} = \Omega^{(\lambda)} V_\lambda$ is contained in $\bigcap_{j=1}^\tau E_j^-|_{V_\lambda}$, where $\tau = \tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$, for $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. The structural sharpness of this containment was left open. We prove the strong sharpness

$$B^{(\lambda)} \cap E_{\tau+1}^-|_{V_\lambda} = \{0\},$$

equivalently, the operator $S_{\tau+1} := T_{\tau+1} + 1$ is *injective* on $B^{(\lambda)}$. The proof is a strong induction on $|\widehat{\lambda}|$ that lifts cleanly through Pillar 1's reduction theorem, with two ingredients: (i) the outer product $\mathcal{O} = \prod_{j=\ell}^{n-2} (T_j + 1)$ is *injective* on $E_{n-1}^+|_{V_\lambda}$ by adjacent-injectivity of the $+q$ -eigenspace chain (May-19 Phase A), and (ii) $S_{\tau+1}$ commutes with every factor of $\mathcal{O}$ and with every Phi-embedding, by the geometric inequality $|j - (\tau + 1)| \geq 2$ that holds throughout the standing regime. The hook base layer ($\widehat{\lambda}$ of length 1) is handled by a separate induction on row length k_h with an explicit 2-block computation at $k_h = 2$.

1 Setup and statement

We adopt the conventions of the Pillar 1 paper [?]. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$. Set $S_j := T_j + 1$ for $1 \leq j \leq n-1$, and write

$$E_j^+ := \ker(T_j - q)|_{V_\lambda} = \text{Im}(S_j|_{V_\lambda}), \quad E_j^- := \ker(S_j)|_{V_\lambda}.$$

For $k \geq 1$, $R'_k := S_{k-1}S_{k-2} \cdots S_1$, with $R'_1 = 1$. For $\nu \vdash N$ with $\ell = \ell(\nu)$,

$$\Omega^{(\nu)} := R'_{\ell+1}^{(\nu)} R'_{\ell+2}^{(\nu)} \cdots R'_N^{(\nu)}, \quad B^{(\nu)} := \Omega^{(\nu)} V_\nu.$$

Throughout, $\widehat{\lambda}$ denotes a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row $\widehat{\lambda}_r \geq 2$, and $\lambda = (\widehat{\lambda}, 1^q) \vdash n$, $n = q + |\widehat{\lambda}|$, $\ell := \ell(\lambda) = \ell(\widehat{\lambda}) + q$. The unfloored threshold is

$$\tau_0(\widehat{\lambda}) := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|, \quad \tau(\widehat{\lambda}) := \max(0, \tau_0(\widehat{\lambda})).$$

Standing hypothesis. $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ (equivalently $\tau_0 \geq 1$).

Theorem 1 (Strong sharpness). *For $\lambda = (\widehat{\lambda}, 1^q)$ with $\widehat{\lambda}$ as above under the standing hypothesis,*

$$B^{(\lambda)} \cap E_{\tau+1}^-|_{V_\lambda} = \{0\},$$

equivalently, $S_{\tau+1}|_{B^{(\lambda)}}$ is injective.

Corollary 2 (Sharpness). $B^{(\lambda)} \not\subseteq E_{\tau+1}^-|_{V_\lambda}$. *Combined with Pillar 1, this gives $\sup\{j : B^{(\lambda)} \subseteq E_j^-|_{V_\lambda}\} = \tau(\widehat{\lambda})$ exactly.*

2 Inputs

We collect three structural inputs.

Proposition 3 (Adjacent-injectivity chain). *For every H_q -module V and every $1 \leq j \leq |\text{ind}| - 2$, $(T_j + 1)|_{E_{j+1}^+(V)}$ is injective with image in $E_j^+(V)$.*

Proof. Image lands in E_j^+ : by $T_j(T_j + 1) = q(T_j + 1)$, so $T_j(T_j + 1)v = q(T_j + 1)v$, i.e., $(T_j + 1)v \in \ker(T_j - q) = E_j^+$. Injectivity: this is the May-19 Phase A adjacent-disjointness statement $E_j^-(V) \cap E_{j+1}^+(V) = 0$, holding for any H_q -module by the $H_q(S_3)$ -irreducible obstruction. So if $v \in E_{j+1}^+$ and $(T_j + 1)v = 0$, then $v \in E_j^- \cap E_{j+1}^+ = 0$. $\square$ $\square$

Lemma 4 (Outer product is injective on E_{n-1}^+). *Set $\mathcal{O} := \prod_{j=\ell}^{n-2} (T_j + 1)$ (product in increasing- j order from left to right). Then $\mathcal{O}|_{E_{n-1}^+|_{V_\lambda}}$ is injective, with image in $E_\ell^+|_{V_\lambda}$.*

Proof. Read the product as $\mathcal{O}v = (T_\ell + 1) \cdots (T_{n-2} + 1)v$, applied right-to-left. For $v \in E_{n-1}^+$, $(T_{n-2} + 1)v \in E_{n-2}^+$ by Proposition ??, and the restriction $(T_{n-2} + 1)|_{E_{n-1}^+}$ is injective. Iterating: at each step $j = n-3, n-4, \dots, \ell$, the input lies in E_{j+1}^+ (by the prior step), $(T_j + 1)$ maps injectively into E_j^+ . After all factors, the image lies in E_ℓ^+ and the composite is injective. $\square$ $\square$

Theorem 5 (Pillar 1 reduction theorem [?]). *Under the standing hypothesis,*

$$B^{(\lambda)} = \mathcal{O} \cdot \sum_{k \in K_{\widehat{\lambda}}} \Phi_{k,*}(B^{(\mu_{k,*})}),$$

where $\Phi_{k,*} : V_{\mu_{k,*}} \hookrightarrow V_\lambda$ is the contributor Phi-embedding (Pillar 1, Lemma 4), T_i -equivariant for $1 \leq i \leq n-3$, and the Phi-images have pairwise disjoint seminormal supports.

Theorem 6 (Pillar 1 r-additivity [?, Thm. A]). $\dim B^{(\lambda)} = \sum_{k \in K_{\widehat{\lambda}}} \dim B^{(\mu_{k,*})}$. *Equivalently, $\mathcal{O} \circ \bigoplus_k \Phi_{k,*}$ is injective on $\bigoplus_k B^{(\mu_{k,*})}$.*

Theorem ?? is in fact strengthened by Lemma ??: $\mathcal{O} \circ \bigoplus_k \Phi_{k,*}$ is injective on the larger space $\bigoplus_k V_{\mu_{k,*}}$, since the Phi-images sit in E_{n-1}^+ with disjoint supports, and $\mathcal{O}$ is injective there.

3 Commutativity geometry

The proof rests on two commutativity facts. Let

$$\delta := \ell - (\tau + 1) = |\widehat{\lambda}| - \ell(\widehat{\lambda}).$$

Lemma 7 (Geometric inequalities). *Under the standing hypothesis,*

- (i) $\delta = |\widehat{\lambda}| - \ell(\widehat{\lambda}) \geq \ell(\widehat{\lambda}) \geq 2$.
- (ii) $n - 1 - (\tau + 1) = 2|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) - 1 \geq 2\ell(\widehat{\lambda}) - 1 \geq 3$.
- (iii) $\tau + 1 \leq \ell - 2$ and $\tau + 1 \leq n - 4$.

Proof. (i) Since every row of $\widehat{\lambda}$ has length ≥ 2 , $|\widehat{\lambda}| \geq 2\ell(\widehat{\lambda})$, so $\delta = |\widehat{\lambda}| - \ell(\widehat{\lambda}) \geq \ell(\widehat{\lambda})$. By hypothesis $\ell(\widehat{\lambda}) \geq 2$. (ii) $n - 1 - (\tau + 1) = (q + |\widehat{\lambda}| - 1) - (q + 2\ell(\widehat{\lambda}) - |\widehat{\lambda}|) = 2|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) - 1 \geq 2 \cdot 2\ell(\widehat{\lambda}) - 2\ell(\widehat{\lambda}) - 1 = 2\ell(\widehat{\lambda}) - 1 \geq 3$. (iii) $\tau + 1 = \ell - \delta \leq \ell - 2$, and $\tau + 1 = n - 1 - (2|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) - 1) \leq n - 4$ from (ii). $\square$ $\square$

Corollary 8 (Commutativity). *Under the standing hypothesis,*

- (a) $S_{\tau+1}$ commutes with every factor of $\mathcal{O}$ (since each factor index $j \geq \ell \geq \tau+3$, so $|j - (\tau+1)| \geq 2$).
- (b) $S_{\tau+1}$ commutes with T_{n-1} (since $|n-1 - (\tau+1)| \geq 3 \geq 2$), hence preserves $E_{n-1}^+|_{V_\lambda}$.
- (c) $S_{\tau+1}$ commutes with each $\Phi_{k,*}$ (since $\tau+1 \leq n-3$, the T_i -equivariance range of $\Phi_{k,*}$).

4 Proof of the main theorem

Proof of Theorem ??. Strong induction on $|\hat{\lambda}|$.

Base case ($|\hat{\lambda}| = 4$, $\hat{\lambda} = (2, 2)$). The unique element of $K_{\hat{\lambda}}$ is $k = 2$ (shrinking corner), and the inner contributor shape is the hook $\mu_{2,*} = (2, 1^q)$, with $\hat{\mu} = (2)$, $\ell(\hat{\mu}) = 1$ (outside the standing hypothesis). By Lemma ?? below, S_q is injective on $B^{(2, 1^q)}$, with $\tau(\hat{\mu}) + 1 = q = \tau(\hat{\lambda}) + 1$ (threshold-matching at the hook layer; see Remark ??). The induction step below applies verbatim with $K_{\hat{\lambda}} = \{2\}$ and IH replaced by Lemma ??.

Inductive step ($|\hat{\lambda}| \geq 5$). Suppose Theorem ?? holds for every $\hat{\mu}$ with $\ell(\hat{\mu}) \geq 2$, every row ≥ 2 , and $|\hat{\mu}| < |\hat{\lambda}|$, and suppose Lemma ?? holds for every hook $(k_h, 1^{m_h})$ with $k_h \geq 2$, $m_h \geq k_h$. Both inputs are available, the former by IH and the latter by Lemma ?? (proved independently in Section ??).

Let $w \in B^{(\lambda)}$ with $S_{\tau+1}w = 0$. We must show $w = 0$.

By the reduction theorem (Theorem ??),

$$w = \mathcal{O} \left(\sum_{k \in K_{\hat{\lambda}}} \Phi_{k,*}(u_k) \right) \quad \text{for some } u_k \in B^{(\mu_{k,*})}.$$

The representation is unique by Theorem ??.

By Corollary ??, $S_{\tau+1}$ commutes with $\mathcal{O}$ and with each $\Phi_{k,*}$. So

$$0 = S_{\tau+1}w = \mathcal{O} \left(\sum_{k \in K_{\hat{\lambda}}} \Phi_{k,*}(S_{\tau+1}u_k) \right).$$

The vector $V := \sum_k \Phi_{k,*}(S_{\tau+1}u_k)$ lies in $\sum_k \Phi_{k,*}(V_{\mu_{k,*}}) \subseteq E_{n-1}^+|_{V_\lambda}$ (Phi-images sit in E_{n-1}^+). By Lemma ??, $\mathcal{O}$ is injective on $E_{n-1}^+|_{V_\lambda}$. From $\mathcal{O}V = 0$ we conclude $V = 0$, i.e.,

$$\sum_{k \in K_{\hat{\lambda}}} \Phi_{k,*}(S_{\tau+1}u_k) = 0.$$

The $\Phi_{k,*}$ -images have pairwise disjoint seminormal supports (Theorem ??). Hence each summand vanishes: $\Phi_{k,*}(S_{\tau+1}u_k) = 0$. Each $\Phi_{k,*}$ is injective, so $S_{\tau+1}u_k = 0$ for each k .

Now apply IH (or, for the base case $\hat{\lambda} = (2, 2)$, Lemma ??) to each $\mu_{k,*}$: by threshold-matching, $\tau(\hat{\mu}_{k,*}) = \tau(\hat{\lambda})$ for every contributor (Pillar 1, Lemma 1). So the IH at index $\mu_{k,*}$ says $S_{\tau+1}|_{B^{(\mu_{k,*})}}$ is injective, hence $u_k = 0$ for each k .

Therefore $w = \mathcal{O} \sum_k \Phi_{k,*}(0) = 0$. □

Remark 9 (Why threshold-matching is needed here). The IH on $\mu_{k,*}$ gives injectivity of $S_{\tau(\hat{\mu}_{k,*})+1}$ on $B^{(\mu_{k,*})}$. The threshold-matching identity (Pillar 1, Lemma 1) guarantees $\tau(\hat{\mu}_{k,*}) = \tau(\hat{\lambda})$ for every contributor (both regular and shrinking cases). So $S_{\tau+1}$ is the operator the IH controls. Without threshold-matching the induction would not close.

5 The hook base layer

Lemma 10 (Hook sharpness). *Let $\mu = (k_h, 1^{m_h})$ with $k_h \geq 2$ and $m_h \geq k_h$. Then $S_{m_h - k_h + 2}$ is injective on $B^{(\mu)}$.*

This is the hook analogue of Theorem ?? with the single-row threshold $\tau_0^{\text{hook}} = m_h - k_h + 1$, $\tau^{\text{hook}} + 1 = m_h - k_h + 2$.

Remark 11. Continuing the unfloored convention $\tau_0(\hat{\mu}) = q + 2\ell(\hat{\lambda})(\hat{\mu}) - 1 - |\hat{\mu}|$ to single-row $\hat{\mu} = (k_h)$ with $q = m_h$ gives $\tau_0 = m_h + 2 - 1 - k_h = m_h - k_h + 1$, matching the hook threshold above. Under the hook-recursion $\mu \rightarrow \mu_{1,*} = (k_h - 1, 1^{m_h - 1})$, τ_0^{hook} is preserved exactly: $(m_h - 1) - (k_h - 1) + 1 = m_h - k_h + 1$.

Proof of Lemma ??. Induction on k_h .

Base case ($k_h = 2$). We have $\mu = (2, 1^{m_h})$ with $m_h \geq 2$, $|\mu| = m_h + 2$, $\ell(\mu) = m_h + 1$, and threshold $m_h - 2 + 2 = m_h$. We must show S_{m_h} is injective on $B^{(\mu)}$.

The chain $\Omega^{(\mu)} = R_{m_h+2}^{(\mu)}$ has a single factor of length $m_h + 1$, and by Phase A

$$B^{(\mu)} = R_{m_h+2}^{(\mu)} V_\mu = E_{m_h+1}^+ |_{V_\mu}.$$

The removable cells of μ are $(1, 2)$ and $(m_h + 1, 1)$; the unique non-column-adjacent cell pair for $\{m_h + 1, m_h + 2\}$ is the pair $\{(1, 2), (m_h + 1, 1)\}$. Hence $\dim E_{m_h+1}^+ |_{V_\mu} = 1$, spanned by

$$w_0 := v_{T^+} + \beta v_{T^-},$$

where T^+ places $m_h + 1$ at $(1, 2)$ and $m_h + 2$ at $(m_h + 1, 1)$; T^- swaps the two; and $\beta \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at axial distance $1 - (1 - m_h) = m_h$.

In T^+ : entry m_h sits at $(m_h, 1)$, entry $m_h + 1$ at $(1, 2)$. These cells lie in different rows and different columns, so the T_{m_h} -block at T^+ is a non-degenerate 2×2 Hoefsmit block. In particular, $S_{m_h} v_{T^+} = (a + 1)v_{T^+} + b v_{s_{m_h} T^+}$ with $b \neq 0$ (off-diagonal Hoefsmit entry at axial distance $1 - (1 - m_h) = m_h \neq \pm 1$ is nonzero in $\mathbb{Q}(q)$).

In T^- : entry m_h at $(m_h, 1)$, entry $m_h + 1$ at $(m_h + 1, 1)$ — same column. So $S_{m_h} v_{T^-} = 0$.

Combining:

$$S_{m_h} w_0 = (a + 1)v_{T^+} + b v_{s_{m_h} T^+}.$$

The vector $v_{s_{m_h} T^+}$ is a basis vector outside $\text{supp}(B^{(\mu)}) = \{T^+, T^-\}$ (the SYT $s_{m_h} T^+$ has m_h at $(1, 2)$, so its prefix length is $m_h - 1 < m_h$; it does not appear in $\text{supp}(B^{(\mu)})$, in fact it does not appear in the support of $E_{m_h+1}^+ |_{V_\mu}$ at all, but as a basis vector in V_μ). Therefore $S_{m_h} w_0$ has a nonzero $v_{s_{m_h} T^+}$ -coefficient $b \neq 0$, so $S_{m_h} w_0 \neq 0$. Since $B^{(\mu)}$ is the line through w_0 , this proves injectivity of $S_{m_h} |_{B^{(\mu)}}$.

Inductive step ($k_h \geq 3$). We invoke the hook analogue of the Pillar 1 reduction theorem, which we record as Lemma ?? below.

By Lemma ??,

$$B^{(\mu)} = \mathcal{O}^{(\mu)} \cdot \Phi_{1,*}^{(\mu)}(B^{(\mu_{1,*})}),$$

where $\mathcal{O}^{(\mu)} = \prod_{j=m_h+1}^{m_h+k_h-2} (T_j + 1)$, $\mu_{1,*} = (k_h - 1, 1^{m_h - 1})$, and $\Phi_{1,*}^{(\mu)}$ is the contributor embedding.

Suppose $S_{m_h - k_h + 2} w = 0$ for some $w \in B^{(\mu)}$. Write $w = \mathcal{O}^{(\mu)} \Phi_{1,*}^{(\mu)}(u_1)$ uniquely with $u_1 \in B^{(\mu_{1,*})}$. We must show $u_1 = 0$.

Geometric inequalities for the hook. With $\ell^{(\mu)} = m_h + 1$ and $\tau^{\text{hook}} + 1 = m_h - k_h + 2$:

$$\ell^{(\mu)} - (\tau^{\text{hook}} + 1) = (m_h + 1) - (m_h - k_h + 2) = k_h - 1 \geq 2,$$

and $|\mu| - 1 - (\tau^{\text{hook}} + 1) = (k_h + m_h - 1) - (m_h - k_h + 2) = 2k_h - 3 \geq 3$. Hence:

- $S_{m_h - k_h + 2}$ commutes with every factor of $\mathcal{O}^{(\mu)}$ (each factor index $j \geq m_h + 1 \geq (m_h - k_h + 2) + 2$).
- $S_{m_h - k_h + 2}$ commutes with $T_{|\mu| - 1}$, hence preserves $E_{|\mu| - 1}^+|_{V_\mu}$.
- $S_{m_h - k_h + 2}$ commutes with $\Phi_{1,*}^{(\mu)}$ (since $m_h - k_h + 2 \leq |\mu| - 3 = k_h + m_h - 3$ iff $2k_h \geq 5$, which holds for $k_h \geq 3$).

The same argument as in the main theorem proceeds:

$$0 = S_{m_h - k_h + 2}w = \mathcal{O}^{(\mu)}\Phi_{1,*}^{(\mu)}(S_{m_h - k_h + 2}u_1).$$

The vector $\Phi_{1,*}^{(\mu)}(S_{m_h - k_h + 2}u_1)$ lies in $E_{|\mu| - 1}^+|_{V_\mu}$, on which $\mathcal{O}^{(\mu)}$ is injective (Lemma ?? applied at the μ level — the proof is identical, since adjacent-injectivity is shape-agnostic). So $\Phi_{1,*}^{(\mu)}(S_{m_h - k_h + 2}u_1) = 0$, and injectivity of $\Phi_{1,*}^{(\mu)}$ gives $S_{m_h - k_h + 2}u_1 = 0$.

The inner threshold matches (Remark ??), and the IH (induction on k_h , since $k_h - 1 < k_h$ and $m_h - 1 \geq k_h - 1$ when $m_h \geq k_h \geq 3$) gives $S_{m_h - k_h + 2}$ injective on $B^{(\mu_{1,*})}$. So $u_1 = 0$, hence $w = 0$. $\square$

Lemma 12 (Hook reduction theorem). *For $\mu = (k_h, 1^{m_h})$ with $k_h \geq 3$ and $m_h \geq k_h$,*

$$B^{(\mu)} = \mathcal{O}^{(\mu)} \cdot \Phi_{1,*}^{(\mu)}(B^{(\mu_{1,*})}),$$

where $\mathcal{O}^{(\mu)} = \prod_{j=m_h+1}^{m_h+k_h-2} (T_j + 1)$, $\mu_{1,*} = (k_h - 1, 1^{m_h-1})$, and $\Phi_{1,*}^{(\mu)}$ is the Phi-embedding for the corner pair $\{(1, k_h), (m_h + 1, 1)\}$.

Proof. The proof is identical to the Pillar 1 reduction theorem, specialised to hooks. We sketch the differences.

Phi-decomposition of $E_{|\mu| - 1}^+|_{V_\mu}$. For hook $\mu = (k_h, 1^{m_h})$, the removable corners of μ are $(1, k_h)$ and $(m_h + 1, 1)$. The cell pairs for $\{|\mu| - 1, |\mu|\}$ are: (i) the corner pair $\{(1, k_h), (m_h + 1, 1)\}$, giving the contributor $\Phi_{1,*}^{(\mu)}$; (ii) the same-row pair $\{(1, k_h - 1), (1, k_h)\}$, giving the vanisher $\Phi_h^{(\mu)}$ when $k_h \geq 3$ (the condition $\hat{\lambda}_1 \geq 3$ is satisfied; the condition $\hat{\lambda}_1 \geq \hat{\lambda}_2 + 2$ collapses since there is no $\hat{\lambda}_2$ for single-row [interpretas automatic]). There are no corner – pair vanishers since $\hat{\lambda}$ has only one corner.

Vanisher kill. For the same-row vanisher $\Phi_h^{(\mu)}$ at the row-1 pair, the inner shape is $(k_h - 2, 1^{m_h})$ if $k_h \geq 4$, or (1^{m_h+1}) (a column) if $k_h = 3$. In either case the hook column-1 theorem [?, Thm. 4] (or trivially for columns, where $B = V_\nu \subseteq E_1^-$ since every SYT has 1, 2 in the same column) gives $B^{(\nu)} \subseteq E_1^-|_{V_\nu}$. The inner pull-through provides an extra $R_\ell^{(\nu)}$ factor whose rightmost component is S_1 , which kills E_1^- . So the vanisher contribution vanishes.

Contributor. For $\mu_{1,*} = (k_h - 1, 1^{m_h-1})$, $\ell(\mu_{1,*}) = m_h$, so the Q -product after inner pull-through has factor indices $\ell^{(\mu)} = m_h + 1$ through $|\mu_{1,*}| = k_h + m_h - 2$, which is exactly the range $\ell(\mu_{1,*}) + 1, \dots, |\mu_{1,*}|$ defining $\Omega^{(\mu_{1,*})}$. Hence the contributor yields $\Phi_{1,*}^{(\mu)}(\Omega^{(\mu_{1,*})}V_{\mu_{1,*}}) = \Phi_{1,*}^{(\mu)}(B^{(\mu_{1,*})})$ after the outer $\mathcal{O}^{(\mu)}$ factors. $\square$

6 Computational verification

We verify Theorem ?? by computing $\dim B^{(\lambda)}$ and $\dim S_{\tau+1}(B^{(\lambda)})$ for each test shape, using a sparse Hoefsmit-basis implementation (working mod $P = 100003$ at $Q = 11$). Sharpness in the strong form is the assertion $\dim S_{\tau+1}(B) = \dim B$ (full rank of $S_{\tau+1}$ on B).

Table 1: Verification of strong sharpness on multi-row shapes $\lambda = (\widehat{\lambda}, 1^q)$ and on hooks $\mu = (k_h, 1^{m_h})$.

shape	$\ell(\widehat{\lambda})$	n	τ	$\dim V$	$\dim B$	$\dim S_{\tau+1}(B)$
$(2, 2, 1^2)$	2	6	1	9	1	1
$(2, 2, 1^3)$	2	7	2	14	1	1
$(2, 2, 1^4)$	2	8	3	20	1	1
$(3, 2, 1^3)$	2	8	1	64	2	2
$(3, 2, 1^4)$	2	9	2	105	2	2
$(2, 2, 2, 1^2)$	3	8	1	28	1	1
$(2, 2, 2, 1^3)$	3	9	2	48	1	1
$(3, 2, 2, 1^3)$	3	10	1	315	3	3
$(2, 1^2)$	—	4	1	3	1	1
$(2, 1^3)$	—	5	2	4	1	1
$(3, 1^3)$	—	6	1	10	1	1
$(3, 1^5)$	—	8	3	21	1	1
$(4, 1^4)$	—	8	1	35	1	1
$(4, 1^7)$	—	11	4	120	1	1

The full rank $\dim S_{\tau+1}(B) = \dim B$ holds in every case tested.

7 Discussion

7.1 What this proves

Sharpness, finally. Theorem ?? closes the structural sharpness gap left open by the Pillar 1 paper, completing the characterisation of the depth of the column-1 chain image:

$$\sup\{j : B^{(\lambda)} \subseteq E_j^-\} = \tau(\widehat{\lambda}).$$

Strong form. The conclusion is a strict strengthening of the conjectured $B \not\subseteq E_{\tau+1}^-$: it shows $S_{\tau+1}$ is *injective* on B , i.e., $B \cap E_{\tau+1}^- = 0$. This places B as a complement to $E_{\tau+1}^-$ inside V_λ , in the sense $B \oplus E_{\tau+1}^- \hookrightarrow V_\lambda$.

7.2 The role of adjacent-injectivity

The heart of the proof is Lemma ??: the outer product $\mathcal{O} = \prod_{j=\ell}^{n-2} (T_j + 1)$ is injective on $E_{n-1}^+|_{V_\lambda}$. This statement is module-agnostic — it depends only on the Hecke quadratic relation and the

adjacent-disjointness $E_j^- \cap E_{j+1}^+ = 0$ established in May-19. It is striking that the same fact that drives r-additivity [?] also drives sharpness, via the simple observation that $S_{\tau+1}$ preserves E_{n-1}^+ .

7.3 Connection to trace-vanishing

Combined with Pillar 1, the present theorem says:

$$B^{(\lambda)} \subseteq \bigcap_{j=1}^{\tau} E_j^-, \quad B^{(\lambda)} \cap E_{\tau+1}^- = 0.$$

On B , T_j acts as -1 for $j \leq \tau$ and has a nonzero $+q$ component for $j = \tau + 1$. In particular, the trace $\text{tr}_B(T_1 T_2 \cdots T_{\tau+1}) = (-1)^\tau \cdot \text{tr}_B(T_{\tau+1})$, where $T_{\tau+1}$ restricted to B has at least one $+q$ eigenvalue. The exact eigenvalue spectrum of $T_{\tau+1}|_B$ is a finer invariant.

7.4 Limitations

Non-stable regime. For $q < |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ (the non-stable regime), neither Pillar 1 nor the present theorem applies. The sharpness picture there is expected to mirror the May-14 non-stable/JM phenomena and is not addressed here.

8 Status

Established.

- Theorem ?? : $B^{(\lambda)} \cap E_{\tau+1}^- = 0$ for all $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$, every row ≥ 2 , $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.
- Lemma ?? : hook sharpness, base layer of the recursion.
- Verification on 14 datapoints (8 multi-row, 6 hook).

Open.

- Eigenvalue spectrum of $T_{\tau+1}|_{B^{(\lambda)}}$.
- Sharpness in the non-stable regime $q < |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.
- Categorical lift: an analogous statement on the triply-graded HOMFLY / Hochschild homology of the Rouquier complex, per `connections/2026-05-16-categorical-home-consolidates.md`.

References

- [1] Clio, *Extended sign-kill for general multi-row $\widehat{\lambda}$: the meta-theorem at arbitrary $\ell(\widehat{\lambda}) \geq 2$* , 15 May 2026. `~/projects/proofs/2026-05-15-multi-row-sign-kill-meta.tex`.
- [2] Clio, *r-additivity and the SYT formula*, 13 May 2026. `~/projects/proofs/2026-05-13-r-additivity-and-syt-formula.tex`.
- [3] Clio, *Phase A endpoint and adjacent-injectivity*, 19 May 2026 (earlier paper number; cited by Pillar 1). `~/projects/proofs/2026-05-19-phase-a-general.tex`.